SPACE GEOMETRY*

14–1 INTRODUCTION

In this chapter we outline the geometry of space, that is, three-dimensional Euclidean space. Sometimes this study is called *solid* geometry. Just as plane geometry is an abstraction derived from our experience with flat surfaces, so solid geometry is an abstraction derived from our ordinary perception of objects in space.

In this chapter you will find many figures and many theorems and definitions, but very few proofs. Proofs are omitted because this chapter is intended to be only an introduction to the subject. Nevertheless, if you study the theorems in the order they occur, you will find that their arrangement is logical because they can be proved in that order. Furthermore, most of them have fairly simple proofs. To get an understanding of the *facts* of solid geometry, it will be sufficient to read over the theorems and be sure that you know what they say.

* This chapter is divided into three parts. For Part A (Section 14–3), Chapters 1 through 3 are prerequisites. For Part B (Sections 14–4 and 14–5), Chapters 4 through 8 are prerequisites. For Part C (Sections 14–6 through 14–9), Chapters 9 through 12 are prerequisites.

14–2 GEOMETRIC OBJECTS

As in plane geometry, we shall be concerned mostly with simple sets of points that you have already encountered, namely, planes, spheres, pyramids, cubes, prisms, etc., as in Fig. 14–1. More complicated sets of points, such as those in Fig. 14–2, will occur to you. As before, any set of points may be studied. The drawings in Fig. 14–2 represent sets of points that do not lie in a plane.

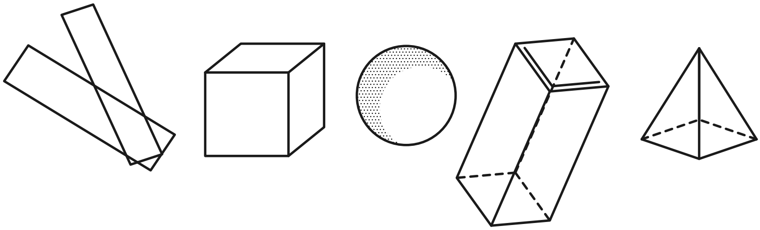

FIGURE 14–1

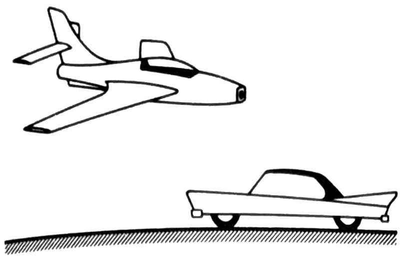

FIGURE 14–2

Exercise Group 14–1

1. Draw figures of several three-dimensional sets of points.
2. Sketch a polyhedron.
3. What might be meant by parallel planes? perpendicular planes?

A. LINES, PLANES, AND BETWEENNESS IN SPACE

14–3 THE POSTULATES

It is perhaps somewhat surprising that most of the essential difficulties in geometry already occur in the plane, so that extension to space geometry requires no fundamentally new concepts. As a matter of fact, very few more postulates are needed than for the plane, indeed only four more.

Postulate VII–1. There is a set of points called *space*. Certain subsets of these points are called planes. There are at least four points that are not in any one plane.

Postulate VII–2. There is exactly one plane containing any three noncollinear points (Fig. 14–3).

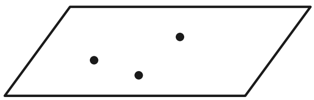

FIGURE 14–3

Postulate VII–3. Each plane satisfies all the postulates for plane geometry. In particular, the line joining two points of a plane lies in that plane. Furthermore, the postulates for congruence of segments and angles apply whether or not the segments and angles are in the same plane.

Postulate VII–4. Each plane separates space. By this we mean that the set of points of space not on the plane consists of two subsets, called the sides of the plane, with the following properties. If two points are in the same subset, then the segment joining them does not meet the plane, but if the two points are in different subsets, the segment joining them does intersect the plane.

Postulate VII–1 asserts that no plane fills up all of space, so that we have a geometry of more than two dimensions (Fig. 14–4).

Postulate VII–2 gives us a simple requirement for fixing a definite plane.

Postulate VII–3 states that our planes are as we want them to be from our study of plane geometry. In addition, this postulate relates congruent lines and angles in different planes.

Postulate VII–4 expresses the familiar fact that a wall between neighbors separates them.

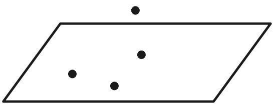

FIGURE 14–4

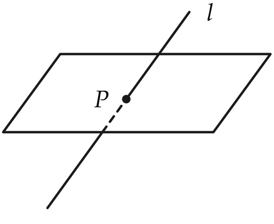

FIGURE 14–5

Exercise 14–2

Restate Postulate Group VII in your own words.

From these postulates and the first two postulate groups for plane geometry, we can prove a number of theorems concerning intersections of lines and planes.

Theorem 14–1. *If a line l does not lie in a plane π, and has at least one point in π, then l and π have exactly one point in common (Fig. 14–5).*

For the rest of the chapter, proofs will be left out, but we include a proof of Theorem 14–1 as a guide to start you out correctly.

Proof. There is, by assumption, at least one point P in both l and π. Suppose that there were a second point Q. Then, by Postulate VII–3, the line $l = PQ$ would lie in the plane π. But we have assumed that l is not in π. Hence P is the only point in both l and π.

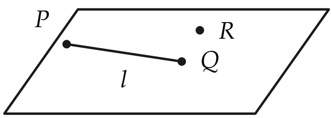

FIGURE 14–6

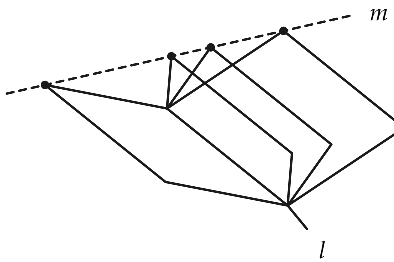

FIGURE 14–7

Theorem 14–2. *There is a unique plane containing a given line and a point not on the line (Fig. 14–6).*

Theorem 14–3. *There are infinitely many planes containing a given line.*

Figure 14–7 may suggest a proof.

Theorem 14–4. *There is a unique plane containing two intersecting lines (Fig. 14–8).*

Theorem 14–5. *If two distinct planes have at least one point in common, they have precisely a line in common (Fig. 14–9).*

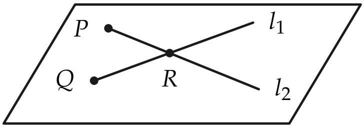

FIGURE 14–8

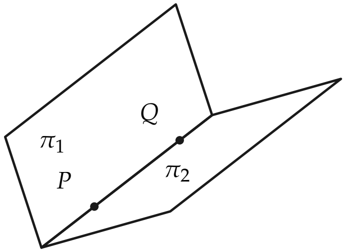

FIGURE 14–9

Definition 14–1. If two planes π_1 and π_2 intersect in a line l (Fig. 14–10), let π_1^+ and π_2^+ be half planes of π_1 and π_2, determined by l. The set of all points in π_1^+ and π_2^+ is called a *dihedral angle*. The line l is called the *edge*, and π_1^+ and π_2^+ are called the *faces*, of the dihedral angle.

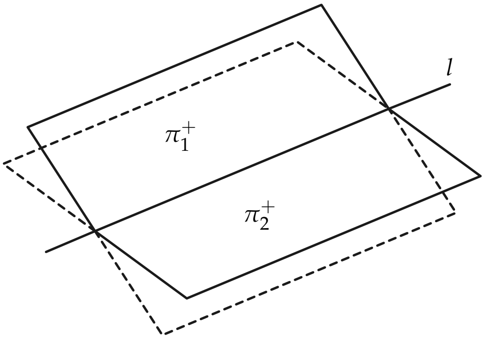

FIGURE 14–10

Definition 14–2. The interior of the dihedral angle with faces π_1^+ and π_2^+ is the set of all points of space on the same side of π_2 as π_1^+ and on the same side of π_1 as π_2^+.

Exercise Group 14–3

1. Give a precise definition for a tetrahedron (Fig. 14–11). The tetrahedron is usually considered to be just the surface. What do you define the interior to be?

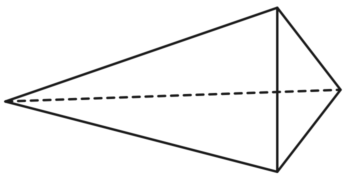

FIGURE 14–11

2. Give a definition for a polyhedron (surface). See Fig. 14–12.

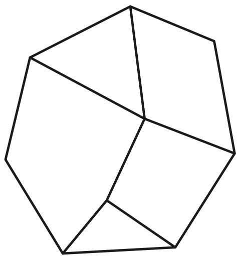

FIGURE 14–12

3. Give a precise definition for a half plane.

B. CONGRUENCE IN SPACE

14–4 PERPENDICULAR LINES AND PLANES

Before giving some theorems, we observe that all right angles are congruent regardless of whether they are in the same plane. We know this because our postulates for angle congruence state that, given an $\angle AOB$ in a plane π and a ray r' from O' in a plane π', there is exactly one ray r'' from O' (on a given side of the line of r' in the plane π') that forms with r' an angle congruent to $\angle AOB$. See Fig. 14–13.

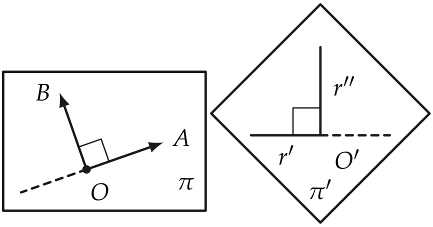

FIGURE 14–13

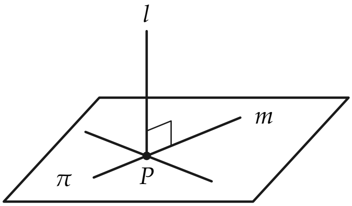

FIGURE 14–14

Definition 14–3. A line l is perpendicular to a plane π if l intersects π at a point P and l is perpendicular to every line in π that passes through P (Fig. 14–14).

Theorem 14–6. *If a line l is perpendicular to two intersecting lines, then it is perpendicular to the plane containing the lines (Fig. 14–15).*

[*Hint:* Choose A and A' such that $AP \cong A'P$. Choose B and C. Then prove that $\triangle ABC \cong \triangle A'BC$. Then prove that $\triangle ABQ \cong \triangle A'BQ$, so that $AQ \cong A'Q$. This will imply that line $m \perp l$.]

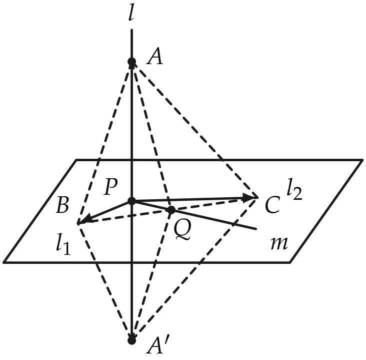

FIGURE 14–15

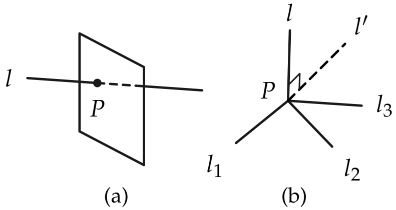

FIGURE 14–16

Theorem 14–7. *There is exactly one plane perpendicular to a line at a point of the line.*

See Fig. 14–16(a).
[*Hint:* Use the previous theorem to show that there is at least one plane. To prove there is only one, it is sufficient to show that if l_1, l_2, and l_3 in Fig. 14–16(b) are distinct lines perpendicular to l at P, then they are in a single plane. Thus, the plane of l_2 and l_3 meets the plane of l and l_1 in a line $l' \perp l$. Since there is only one line in the plane of l and l_1 perpendicular to l at P, we have $l_1 = l'$. Hence l_1, l_2, and l_3 are coplanar.]

Theorem 14–8. *There is exactly one plane perpendicular to a line and containing a point not on the line (Fig. 14–17).*

Exercise Group 14–4

1. State two theorems, analogous to Theorems 14–7 and 14–8, concerning existence of lines perpendicular to planes.
2. Prove the theorems that you stated in Exercise 1.

Theorem 14–9. *If two lines are perpendicular to the same plane, they are parallel (Fig. 14–18).*

[*Hint:* First one must show that the two lines are in one plane.]

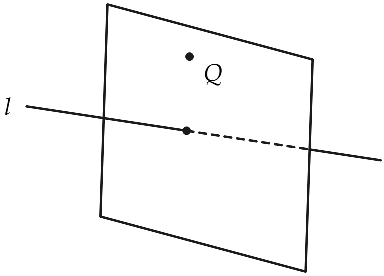

FIGURE 14–17

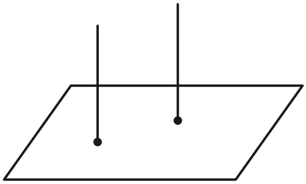

FIGURE 14–18

Definition 14–4. Two lines are called *skew* if they do not lie in one plane. Two planes are called *parallel* if they do not intersect.

Remarks. Two lines in one plane either intersect or are parallel. Two lines in parallel planes need not be parallel. See Fig. 14–19.

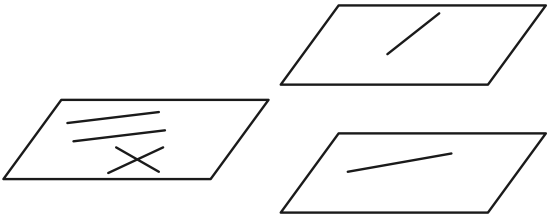

FIGURE 14–19

Theorem 14–10. *Through a point P not in a plane π, there is exactly one plane parallel to π (Fig. 14–20).*

[*Hint:* Choose intersecting lines l_1 and l_2 in π. Then P, with these lines, determines two planes π_1 and π_2, in which there are unique lines l_1' and l_2' parallel to l_1 and l_2, respectively, as in Fig. 14–21. These lines determine a plane parallel to π. The uniqueness is yet to be proved.]

Theorem 14–11. *Two planes perpendicular to the same line are parallel (Fig. 14–22).*

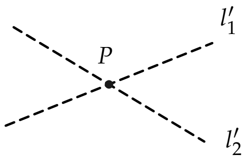

FIGURE 14–20

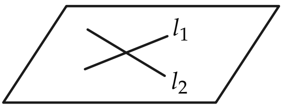

FIGURE 14–21

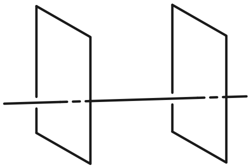

FIGURE 14–22

Theorem 14–12. *If two lines are cut by three parallel planes, the intercepted segments have lengths that are proportional. In Fig. 14–23, we would have $\overline{AB}/\overline{A'B} = \overline{BC}/\overline{B'C'}$.*

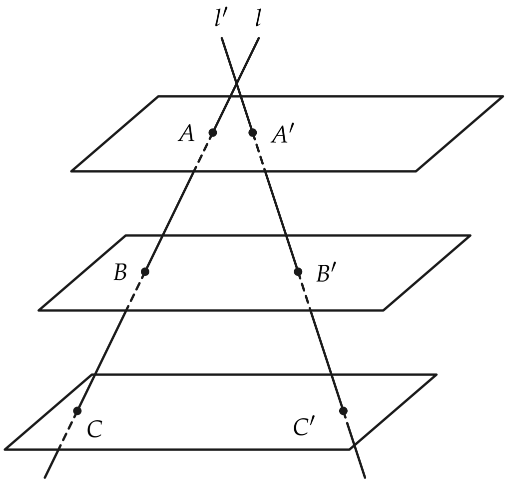

FIGURE 14–23

Exercise Group 14–5

Prove the following theorems.

1. Two planes parallel to one plane are parallel to each other.
2. If a line intersects one of two parallel planes, it intersects the other.
3. If a line is perpendicular to one of two parallel planes, it is perpendicular to the other.
4. If a plane intersects one of two parallel planes, it intersects the other.
5. If one of two parallel lines is perpendicular to a plane π, then the other line is perpendicular to π.

14–5 POLYHEDRA AND POLYHEDRAL ANGLES

Definition 14–5. Let $P_1P_2 \ldots P_n$ be a simple closed polygon in a plane π, and let V be a point not in π, as in Fig. 14–24. The set of points Q on rays from V through all the points P of the polygon is called a polyhedral angle. V is the *vertex*, and $VP_1, VP_2, \ldots VP_n$ are called *edges*. The planes VP_1P_2, $VP_2P_3, \ldots$ contain the *faces* of the angle. The angles $\angle P_1VP_2$, $\angle P_2VP_3, \ldots$ are called *face angles*. The polyhedral angle is called convex if the polygon $P_1 \ldots P_n$ is convex.

Exercise Group 14–6

1. Draw a trihedral angle and a tetrahedral angle. These have three and four faces, respectively.
2. Can a trihedral angle have three right-angled face angles?

Theorem 14–13. *The sum* of two face angles of a trihedral angle is greater than the third face angle (Fig. 14–25).*

[*Hint:* It suffices to prove the theorem in the case in which the third angle is the largest. Suppose that that angle is $\angle AVB$, as in Fig. 14–25. Choose C such that $\angle CAV \cong \angle BAV$. Choose D on AB such that $AC \cong AD$. Now show that $VC \cong VD$ and $CB > DB$. Hence conclude that $\angle CVB > \angle DVB$.]

* The word *sum* is not defined, but the idea is clear.

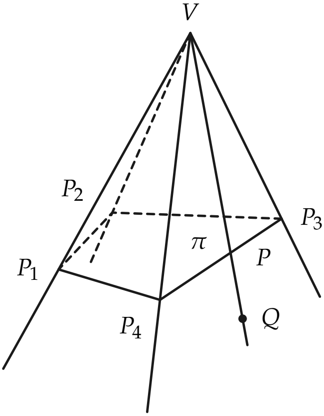

FIGURE 14–24

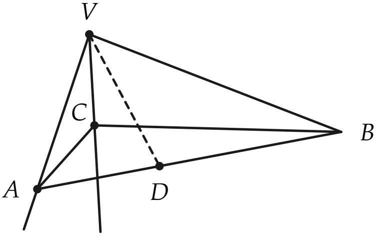

FIGURE 14–25

Theorem 14–14. *The sum of the face angles of a convex polyhedral angle is less than four right angles.*

Exercise 14–7

Illustrate the truth of Theorem 14–14 by constructing a polyhedral angle (Fig. 14–26) of paper and observing that when it is "flattened out" in a plane, it does not "encircle" a point.

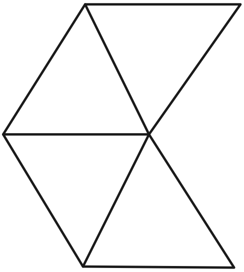

FIGURE 14–26

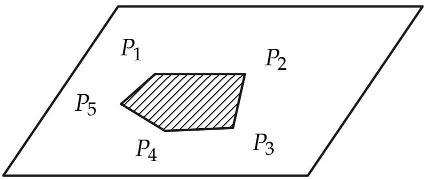

FIGURE 14–27

Definition 14–6. A *polygonal cell* (Fig. 14–27) is the sets of points on, or interior to, a simple plane polygon $P_1P_2 \ldots P_{n-1}$. The edges of the cell are the sides P_1P_2, $P_2P_3, \ldots$ of the polygon.

A *polyhedron* is the set of points on a finite number of polygonal cells joined together in the following way:

(1) Any two cells have either no points in common, exactly one vertex in common, or exactly one edge in common.

(2) Every edge is on precisely two cells.

Exercise 14–8

With Scotch tape and cardboard, construct a number of polyhedra.

Theorem 14–15. *A polyhedron separates space in the following sense. The points not on the polyhedron consist of two sets (called the inside and the outside) such that a polygonal path joining an inside point and an outside point must intersect the polyhedron, and two inside (or two outside) points can be joined by a polygonal path not intersecting the polyhedron.*

Exercise 14–9

Describe the interior of a tetrahedron.

Definition 14–7. A polyhedron is called *convex* if the segment joining any two points of the polyhedron lies either in the polyhedron or its interior. A polyhedron is *regular* if it is convex and all faces are congruent regular polygons.

Exercise 14–10

With Scotch tape and cardboard, construct regular polyhedra with faces that are triangles; squares; pentagons.

Theorem 14–16. *There are but five (except for size) regular polyhedra (Fig. 14–28).*

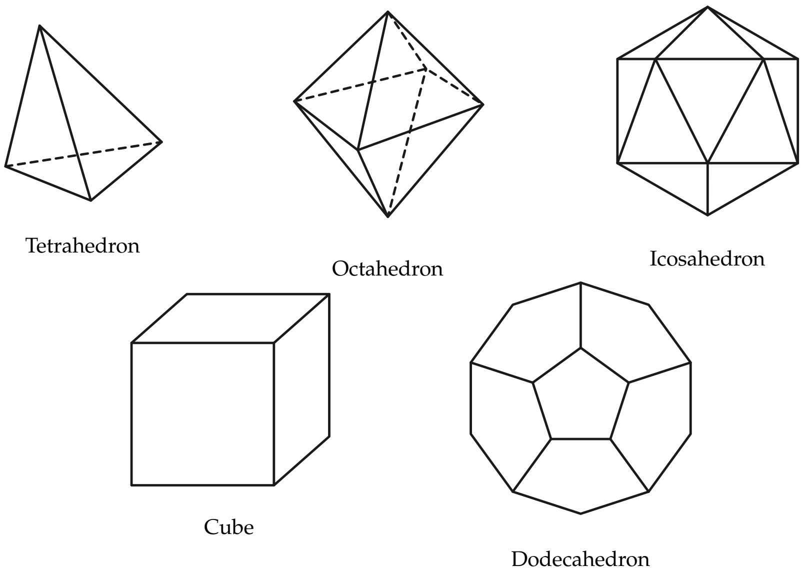

FIGURE 14–28

C. MEASUREMENT IN SPACE

14–6 ANGLES

Theorem 14–17. *Parallel planes intercept a dihedral angle in congruent angles (Fig. 14–29).*

Definition 14–8. The measure of a dihedral angle is equal to the measure of the angle intercepted by a plane perpendicular to the edge of the dihedral angle (Fig. 14–30).

Exercise 14–11

Restate Theorems 14–13 and 14–14, using degrees for measuring angles.

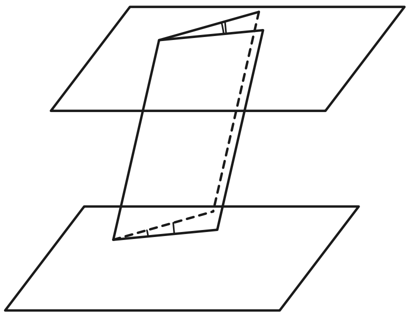

FIGURE 14–29

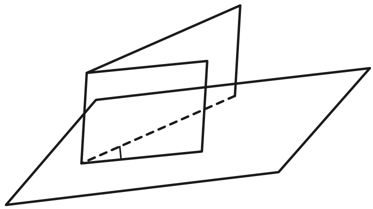

FIGURE 14–30

14–7 AREA

Definition 14–9. The area of a polyhedron is the sum of the areas of its faces.

Exercise Group 14–12

1. A *prism* is a polyhedron two of whose faces are congruent polygons in parallel planes, called bases, and whose remaining edges are all parallel to one another, as in Fig. 14–31. The lateral area is the total area of the faces less the area of the bases. Find a formula for the lateral area of a prism.

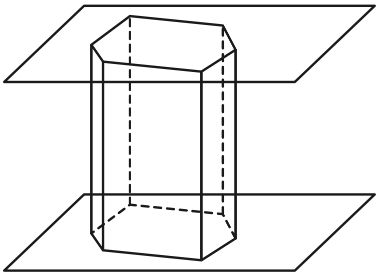

FIGURE 14–31

2. Using Fig. 14–32, define *pyramid* and *regular pyramid*. Find a formula for the lateral area of a regular pyramid.

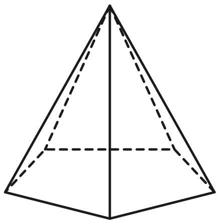

FIGURE 14–32

3. A *parallelepiped* is a special kind of prism. Give a definition that you think fits the drawing in Fig. 14–33. Prove that the opposite faces of a parallelepiped are congruent.

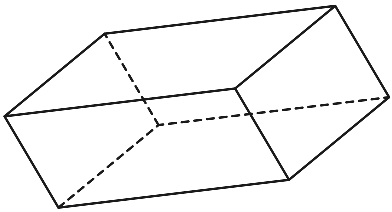

FIGURE 14–33

14–8 VOLUME

Just as area in the plane offers special difficulties, so does volume in space. We content ourselves with listing the theorems and sketching a few "proofs," obtained by drawing figures. Proofs that involve the notion of limit are not presented.

Theorem 14–18. *The volume of a rectangular parallelepiped (box) is equal to the product of the lengths of three concurrent edges. Volume* $= lwh$.

See Fig. 14–34.

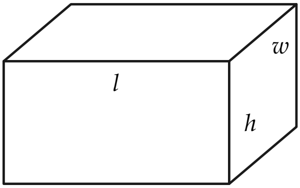

FIGURE 14–34

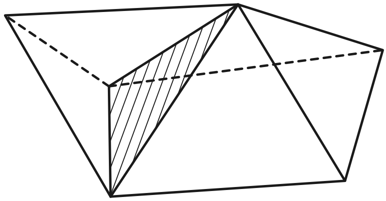

FIGURE 14–35

Theorem 14–19. *If a polyhedron is the union of two other polyhedra, with only a face and no interior points in common, then its volume is the sum of the volumes of the two polyhedra forming it (Fig. 14–35).*

Exercise 14–13

A parallelepiped is shown in Fig. 14–36. Show, by "chopping up" the parallelepiped to form a box, that the following theorem is true.

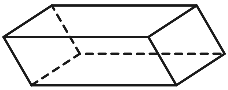

FIGURE 14–36

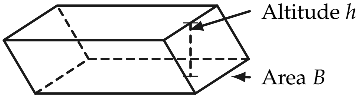

FIGURE 14–37

Theorem 14–20. *The volume of a parallelepiped is equal to the area of a base multiplied by the corresponding altitude (Fig. 14–37).*

The same formula holds for the volume of a prism. To see this, we first prove the theorem for prisms with triangular bases.

Exercise Group 14–14

1. By constructing a parallelepiped of which a triangular prism is half, show that the volume of the prism is equal to the area of a base times the altitude. See Fig. 14–38(a).

FIGURE 14–38

2. Show that the same result holds for an arbitrary prism. See Fig. 14–38(b).

Theorem 14–21. *The volume of a prism is equal to the area of a base times the altitude.*

Volumes of pyramids are somewhat harder to discuss. The following theorem is to be taken without proof.

Theorem 14–22. *If B is the area of the base of a pyramid, then the volume $= \frac{1}{3}Bh$, where h is the altitude (Fig. 14–39).*

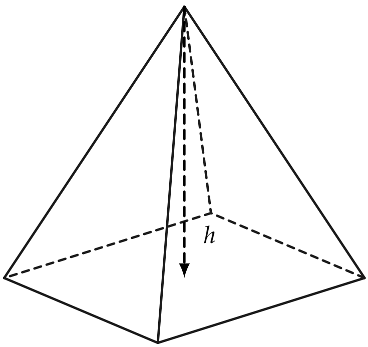

FIGURE 14–39

Exercise Group 14–15

1. Find the volume of a regular pyramid with a base 4 ft square and altitude equal to 4 ft.
2. Find the volume of a regular tetrahedron whose side has length a.
3. Show that if the formula for volume of a pyramid is correct for pyramids with triangular bases, then it is correct for all pyramids.
4. Define *similar polyhedra*, and prove that the volumes of similar polyhedra are proportional to the cubes of corresponding edges. Thus, $V/V' = l^3/l'^3$, as in Fig. 14–40.

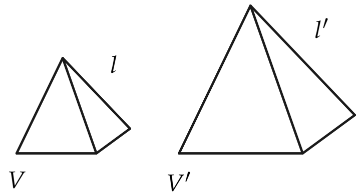

FIGURE 14–40

14–9 CYLINDERS, CONES, AND SPHERES

The familiar shapes in Fig. 14–41 are analogous to prisms, pyramids, and regular polyhedra.

The lateral surface area of cylinders and cones is obtained as a limit of the areas of prisms and pyramids.

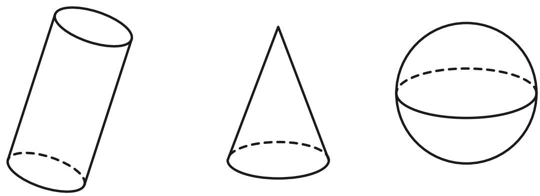

FIGURE 14–41

Exercise Group 14–16

1. Give precise definitions for cylinders, cones, and spheres.
2. Find a formula for the lateral surface of a "regular" cone, called a *right circular cone* (Fig. 14–42).

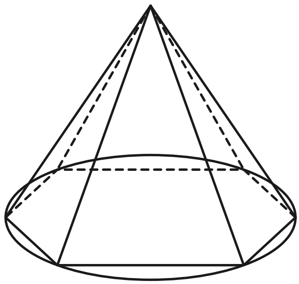

FIGURE 14–42

The surface area of a sphere is harder to discuss. We merely state the result.

Theorem 14–23. *The area of the surface of a sphere of radius r is* $4\pi r^2$.

With the aid of Theorem 14–23, it is not too hard to arrive at the volume of a sphere. Try to prove the following theorem by considering small "pyramids," as shown in Fig. 14–43.

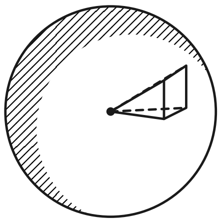

FIGURE 14–43

Theorem 14–24. *The volume of a sphere of radius r is* $\frac{4}{3}\pi r^3$.

Exercise Group 14–17

1. Show that if the radius of a sphere is doubled, the volume is multiplied by a factor of 8.
2. Oranges 2 in. in diameter are 50¢ per dozen. Neglecting skin thickness, how much should one pay for oranges 3 in. in diameter?
3. Prove that if a plane intersects a sphere, then it intersects the sphere in either a circle or one point.
4. What is the formula for the lateral area of a cylinder?